

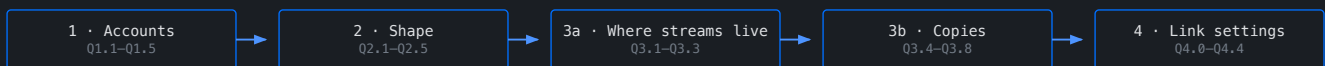
Which NATS setup fits two regions?

Answer the questions in order and follow the arrows. Each answer tells you what to do, and the arrow shows which question comes next. Every answer comes from what the lab measured, unless it says "not tested yet".

How to read it

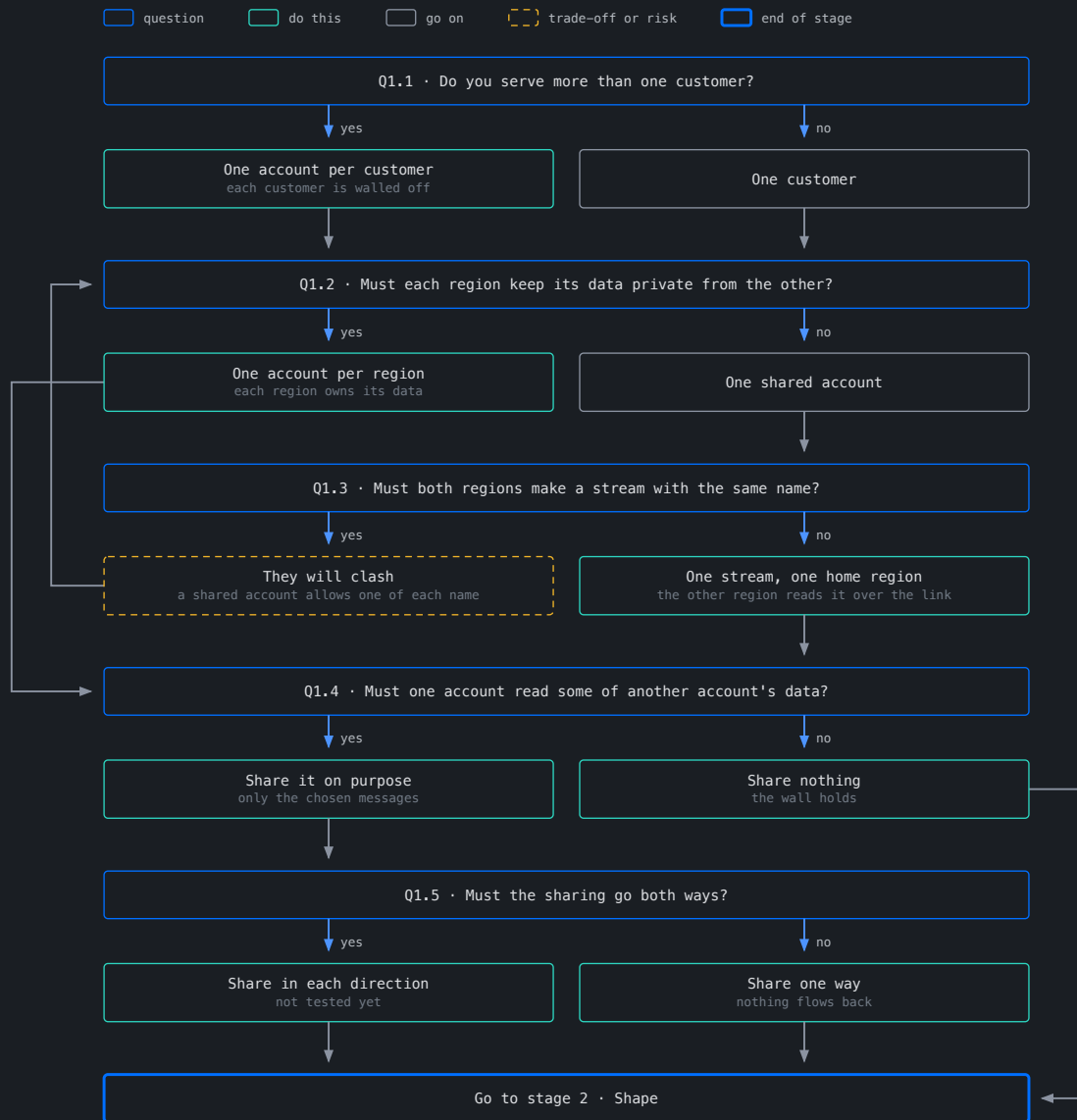
- **Follow the arrows.** Start at the top of each stage. Blue arrows are your yes or no answer. Grey arrows lead to the next question, or around questions you can skip.
- **Answer in order.** Go from stage 1 to stage 4. Ask stages 3a and 3b once for each stream.
- **A region is one data centre.** The tree assumes two regions. "Dark" means a region is down or cut off.
- **A stream is a stored log of messages.** An account is a wall: apps in one account cannot see another account's data unless it is shared on purpose.
- **A vote decides changes.** Creating or changing a stream needs a majority of servers. Sending messages into a stream that already exists needs only that stream's own servers to keep a majority. A stream stored in a dark region stops.
- **The shapes.** T1 = no link. T2 = gateway, one shared system. T3 and T4 = gateway plus a tie-breaker site. T5 = leaf link, each region its own system. T6 = gateway plus leaf.
- **"Not tested yet" is a real gap.** Treat those answers as a guess until the lab measures them.

The five stages



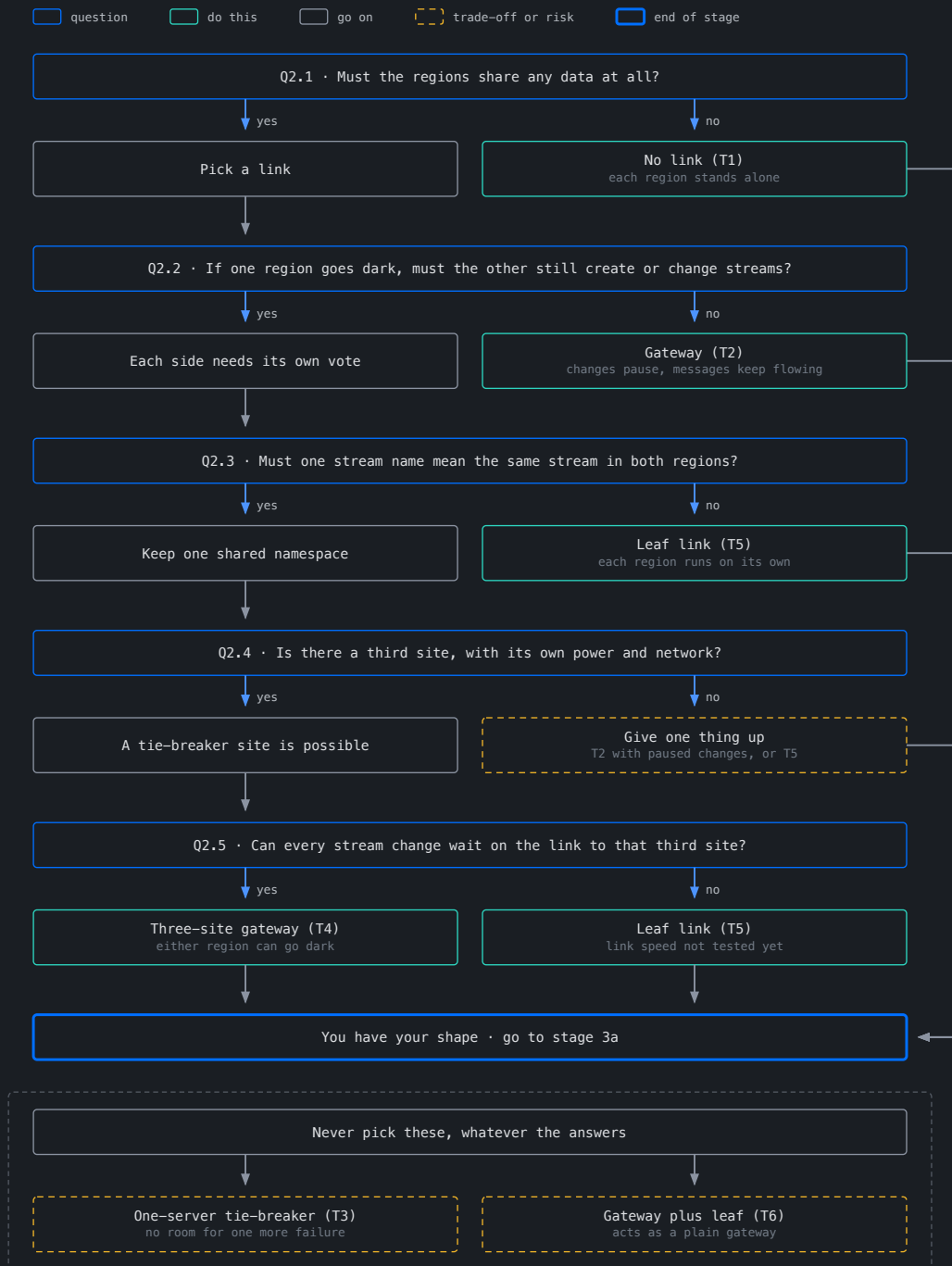
Accounts come first. They decide who can see what. The shape comes next. Then decide, for each stream, where it lives and whether it needs a copy. Link settings come last.

Stage 1 · Accounts — who can see what



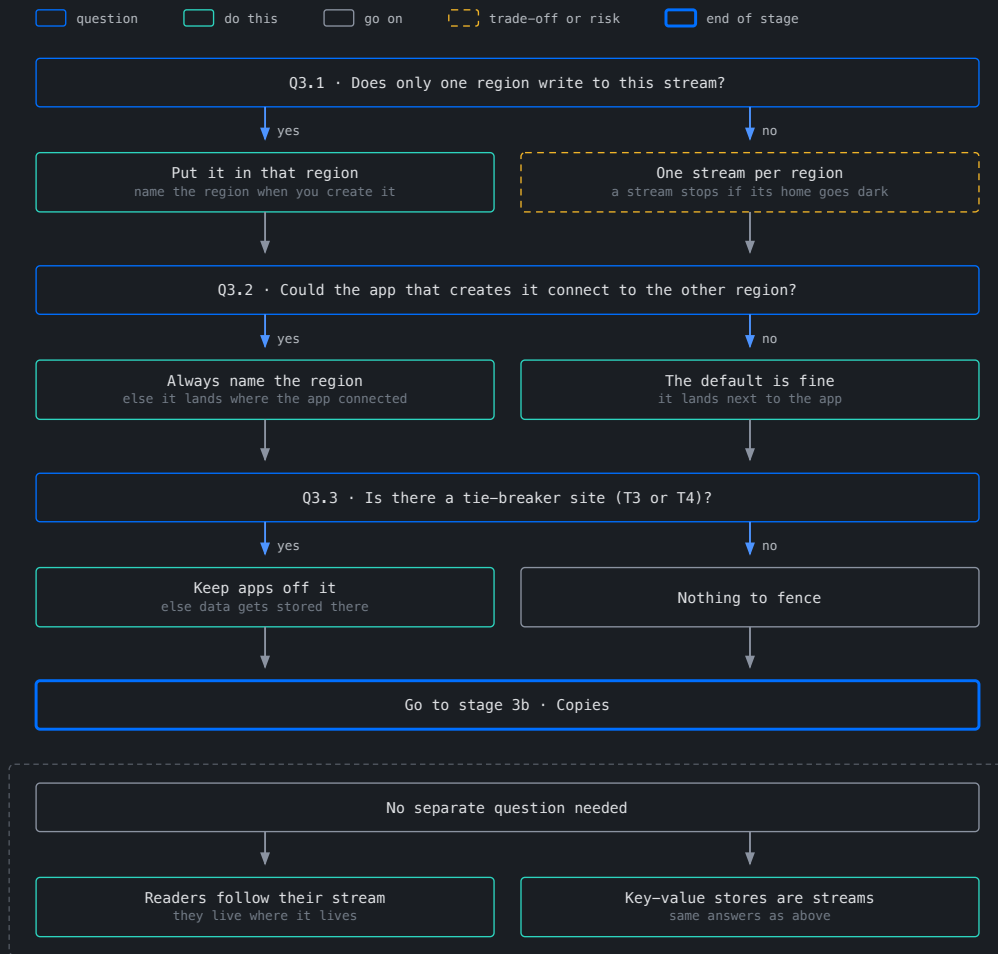
The account is the wall. Accounts decide who can see what. Answer them first. In one shared account, the two regions share one set of stream names, so the same name clashes.

Stage 2 · Shape — how the regions link



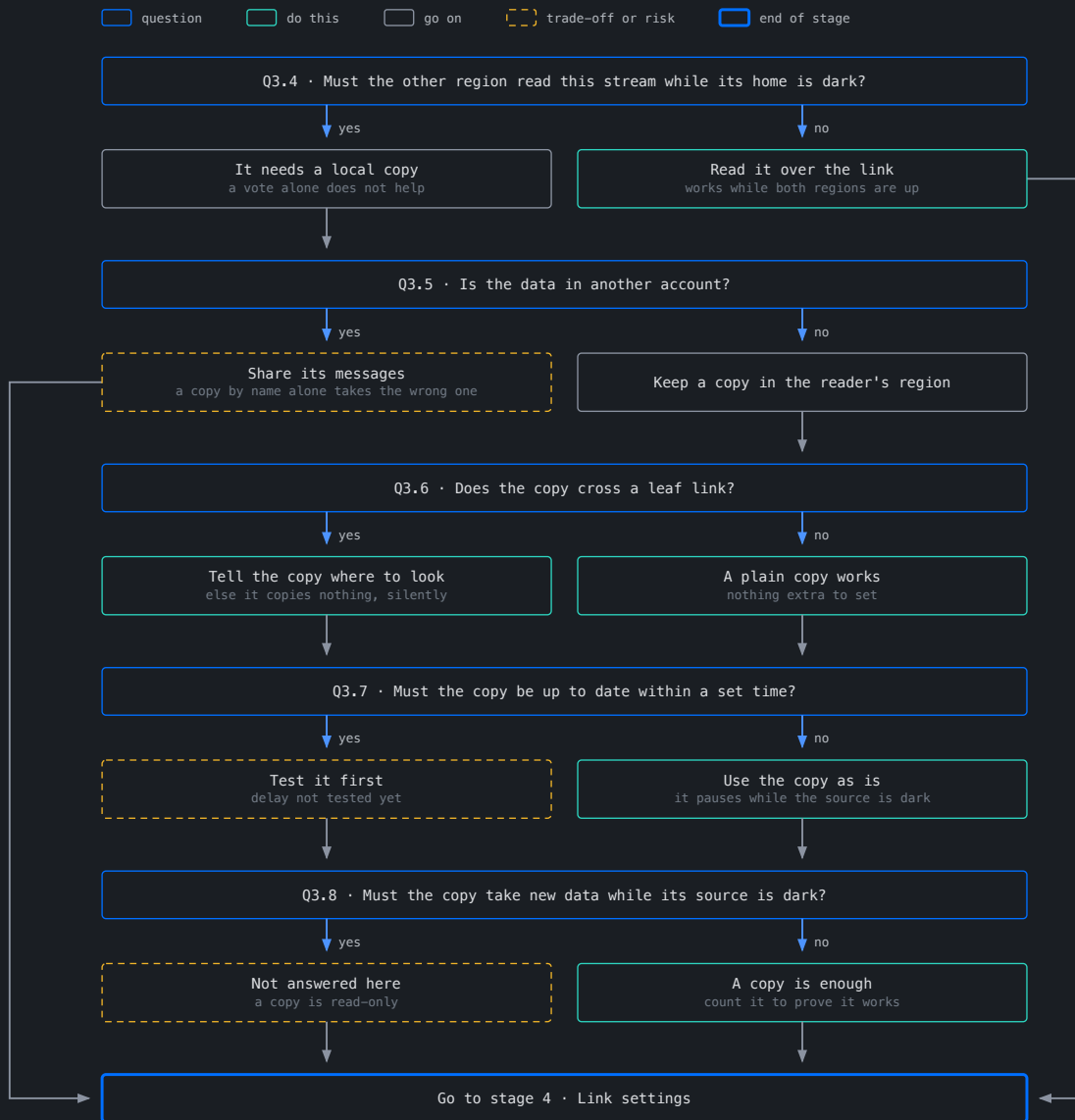
Shape is a vote count. Stream changes need a majority vote. With two regions of equal size, neither side has a majority alone. A third site breaks the tie.

Stage 3a · Where each stream lives — ask once per stream



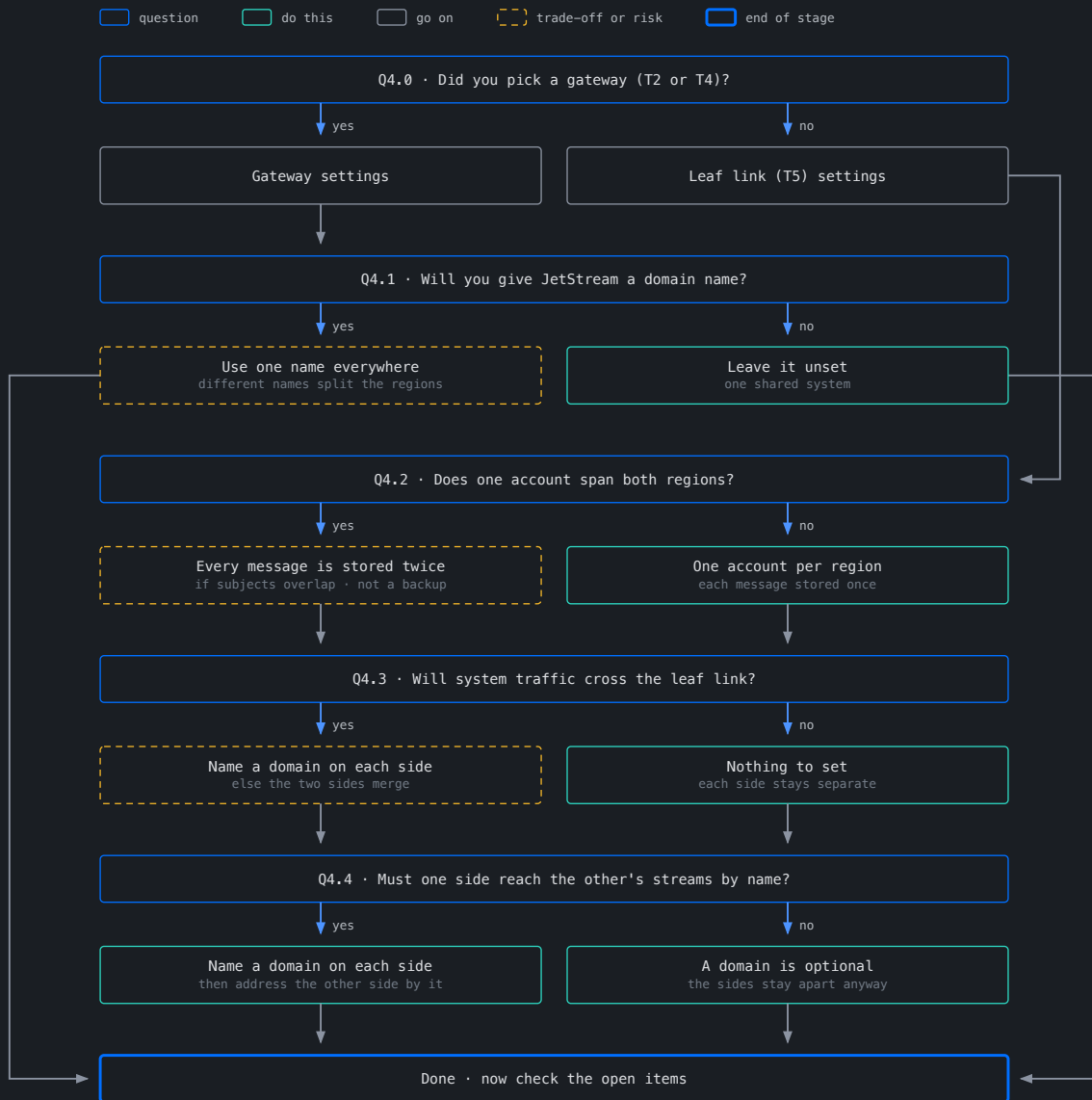
A stream lives where it was created. All its copies stay in that one region. Choose the region on purpose.

Stage 3b · Copies — ask once per stream



A vote keeps the system running, not the data. With a tie-breaker site, the live region can still make changes. It still cannot read streams stored in the dark region (F28). Only a local copy helps. A copy across accounts by stream name alone copies the reader's own stream of that name (E17, E18). NATS documents a configured cross-account copy ([docs](#)); this lab did not test it.

Stage 4 · Link settings — for the shape you picked



A domain is a name for one side's JetStream. It is not a security wall; accounts are. No link (T1)? Skip this stage. On a gateway, use one name or none. On a leaf link, a name on each side keeps the sides apart and lets one reach the other.

Rules for every setup

- **Check the leader twice.** After a failure, the status page can show an old leader for a while. Check again before you act.
- **A named leader is not ready.** The status page can name a new leader before changes work (F6a). Ready means a real change worked. Retry, and record the wait.
- **Count the messages.** A send can report success when nothing was stored.
- **Check every copy by counting.** A wrong or empty copy gives no error.
- **The central hub is optional.** With leaf links, the regions keep working when the hub is gone.
- **A prefix is not proof of origin.** It keeps shared messages apart. Reserve it with publish permissions (E3a).
- **Choose before the data lands.** The one tested way to move between T4 and T5 (stop, change the configs, restart on the same storage) has not shown safe, repeatable operation in two runs, 2026-09-28 and 2026-09-29 (D03-R11, D03-R12). Other ways are untested. It is not proven one-way. Plan the move before you need it.

Not tested yet — measure before you rely on it

- **A real network cut.** Both regions alive but unable to reach each other.
- **Partial loss.** Only some servers in a region fail.
- **Leaf link with a region dark.** Only the hub going dark was tested.
- **Copy delay and link speed.** How far behind a copy runs, and how long a change waits.
- **Placement tags.** Pinning streams to chosen servers.
- **A customer that spans regions.**
- **Two-way sharing.** Only one direction was tested.
- **A configured copy across accounts.** NATS documents it; the lab tested only a copy by name alone.
- **Other ways to move between T4 and T5.** For example, moving to newly built clusters.
- **Why moving T5 to T4 removed ZA copies,** and whether they can be recovered.
- **The hub as the main store.** The hub can hold streams (H8, H11). Carrying both regions under real load, and what its loss costs, is not measured.

Sources

- [Lab report](#), generated by `lab/run-all.sh`
- [Findings deck v1.0](#) · [Pattern cards v0.5](#) · [Decision guide v0.2](#)
- NATS docs: [JetStream across accounts](#) · [leaf nodes and domains](#) · [leaf node remotes](#)

Based on demo 03's lab results, findings deck v1.0 and pattern cards v0.5. Version 1.1, 2026-09-29. Supersedes v1.0.